FACES

Dialogue context

Visual evidence

SFT → format + outcome warm-up

$r^{Fmt} + r^{Out}$

Policy π_θ

Candidates $Y(x)$

for each $y_i \in Y(x)$

Structured candidate y_i

Firstly Vis | Secondly Emo | Thirdly Sit | Fourthly Str | Response

Chain-of-thought (CoT) reasoning spans

Supportive reply

Multi-source semantic rewards

CLIP

Visual evidence

Vis span

CLIP

r^{Vis}

MiniLM

ref. emotion

Emo span

MiniLM

r^{Emo}

MLLM judge

ref. annotations

Sit span

Str span

Response span

MLLM judge

r^{Sit}

r^{Str}

r^{Out}

Five dimension rewards for y_i

Training Only

Non-compensatory gated fusion

structured y_i

Format valid?

Yes

No

$R_i = 0$

r^{Vis} r^{Emo} r^{Sit} r^{Str} r^{Out}

Cognitive mean

$(r^{Vis} + r^{Emo} + r^{Sit})/3$

$\times r^{Str}$ $\times r^{Out}$ R_i

$R_i = \text{mean}(r^{Vis}, r^{Emo}, r^{Sit}) \times r^{Str} \times r^{Out}$

(for format-valid candidates)

Fine-grained token credit

Group rank within $Y(x)$

rank

R_i

positive-only g_i

A > 0: × g_i

A ≤ 0: unchanged

dimension-wise advantages (A)

Vis | Emo | Sit | Str | Out

Clipped GRPO update

earlier: λA

last m: full A

Str span tokens (example)

update policy π_θ (via token-level learning signal)